IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2014

EEE PART II: MEng, BEng and ACGI

Corrected Copy

# ANALOGUE ELECTRONICS 2

Wednesday, 18 June 2014 2:00pm

Time allowed: 2 hours

**There are THREE questions on this paper.**

**ALL questions are compulsory.**
**Question 1 carries 40% of the marks and Questions 2 and 3 carry 30% each.**

**Any special instructions for invigilators and information for candidates are on page 1.**

**Basic expressions for voltage gain, input and output resistance for single stage amplifiers are given on the last page.**

Examiners responsible First Marker(s): T.G. Constandinou
Second Marker(s): C. Toumazou

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1. This question consists of 6 sub-questions. Each item can be answered in a few words or a short paragraph with use of a diagram and/or expression where helpful. Please answer all sub-questions and simplify expressions where possible stating any assumptions made.

a) Explain what is meant by *source degeneration* in MOSFET amplifiers (or emitter degeneration) in BJT amplifiers. What are the advantages/disadvantages of using this technique? [5]

b) Derive expressions (by inspection) for the *voltage gain* of the amplifier circuits shown below (assuming $\lambda \neq 0$, i.e. including $r_O$). [10]

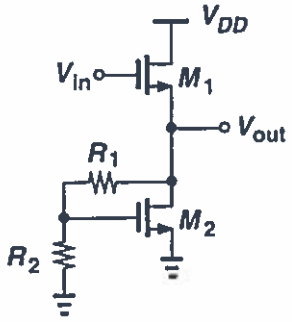



Fig. 1.1(a)

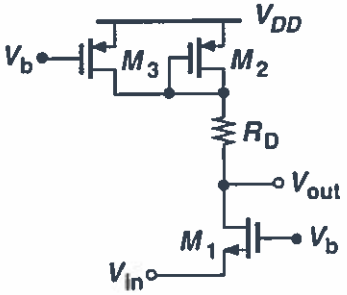



Fig. 1.1(b)

c) Derive expressions (by inspection) for the input or output resistance (as indicated) for the circuits shown below (assuming $V_A < \infty$, i.e. including $r_O$). [10]

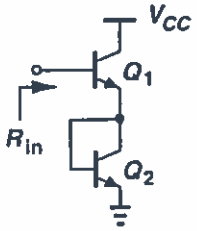



Fig. 1.2(a)

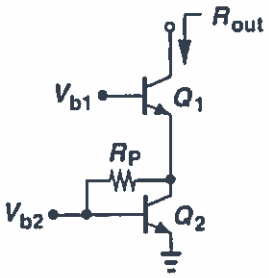



Fig. 1.2(b)

d) Explain how the circuit implementation of a differential amplifier can affect the following characteristics: (i) *output voltage range*, and (ii) *slew rate*. [5]

e) A commercial op-amp (LM741) is quoted an *open-loop gain* of approximately 106dB. What would be the *gain error* if configured in closed-loop as an inverting amplifier for a voltage gain of 200? [5]

f) Describe how *Miller's theorem* can be exploited in op-amp design in improving phase margin. [5]


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2. The circuit shown below is a laser diode driver based on a transconductance amplifier with feedback.

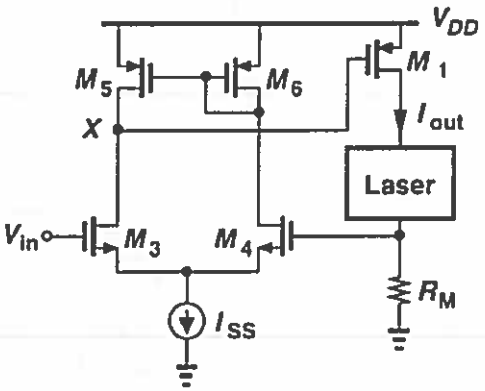



Fig. 2.1

a) Determine the following open loop parameters of the transconductance amplifier (hint: you will need to break the loop):

(i) Open loop gain, [5]

(ii) Open loop output resistance. [5]

b) Determine expressions for the *feedback factor* and "error" signal. [5]

c) Using your answers to (a) and (b), provide an expression for the *closed loop output resistance*. [5]

d) Describe how the circuit designer can maximise the *loop gain* in this circuit. [5]

e) Explain how the resistance of $R_M$ should be selected. Given $V_{DD}$=10V and $I_{out}$=5mA, estimate a value for $R_M$ justifying your estimate. [5]


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3. The circuit shown below is a single stage fully differential amplifier, often referred to as a *folded cascode*.

Use the following parameters: $V_{DD}$=1.8V, $\mu_n C_{ox}$ = 200μA/V$^2$, $V_{THN}$ = 0.4V, $\lambda_N$ = 0.1V$^{-1}$, $\mu_p C_{ox}$ = 100μA/V$^2$, $V_{THP}$ = -0.5V, $\lambda_P$ = 0.2V$^{-1}$, $C_{GS}$=(2/3)(WL)$C_{ox}$, $C_{ox}$=12fF/μm$^2$, $C_{SB}$=$C_{DB}$=0 and $C_{GD}$=$C_0$W, where $C_0$=0.2fF/μm denotes the gate-drain capacitance per unit width.

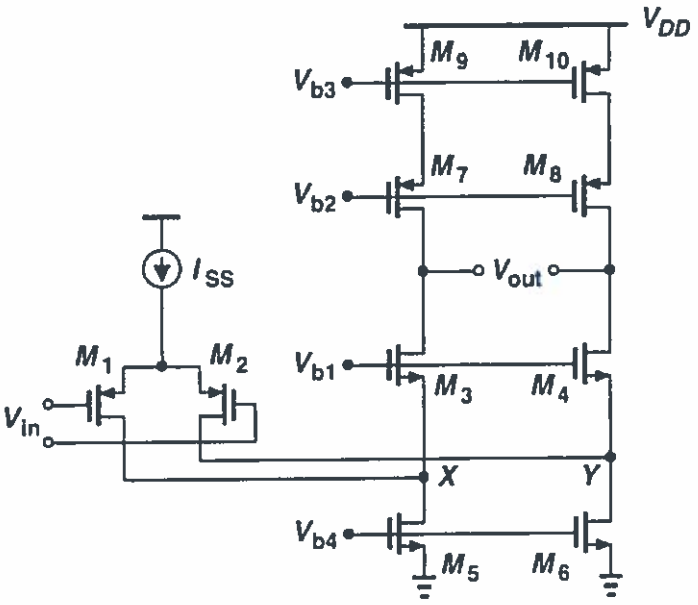



Fig. 3.1

The design parameters are as follows:

| $(W/L)_{M1,2}$ | $(W/L)_{M3,4}$ | $(W/L)_{M5,6}$ | $(W/L)_{M7,8}$ | $(W/L)_{M9,10}$ | $I_{SS}$ | $I_{D3}$, $I_{D4}$ | $I_{D5}$, $I_{D6}$ |
|---|---|---|---|---|---|---|---|
| 100/0.5 | 2/2 | 4/2 | 5/2 | 10/2 | 400μA | 250μA | 50μA |

a) Assuming perfect symmetry, redraw the circuit above to:

(i) Simplify to a single-ended equivalent circuit (i.e. use half circuit approximation). [5]

(ii) Include all the parasitic device capacitances that appear on nodes $X$ and $V_{out}$. [5]

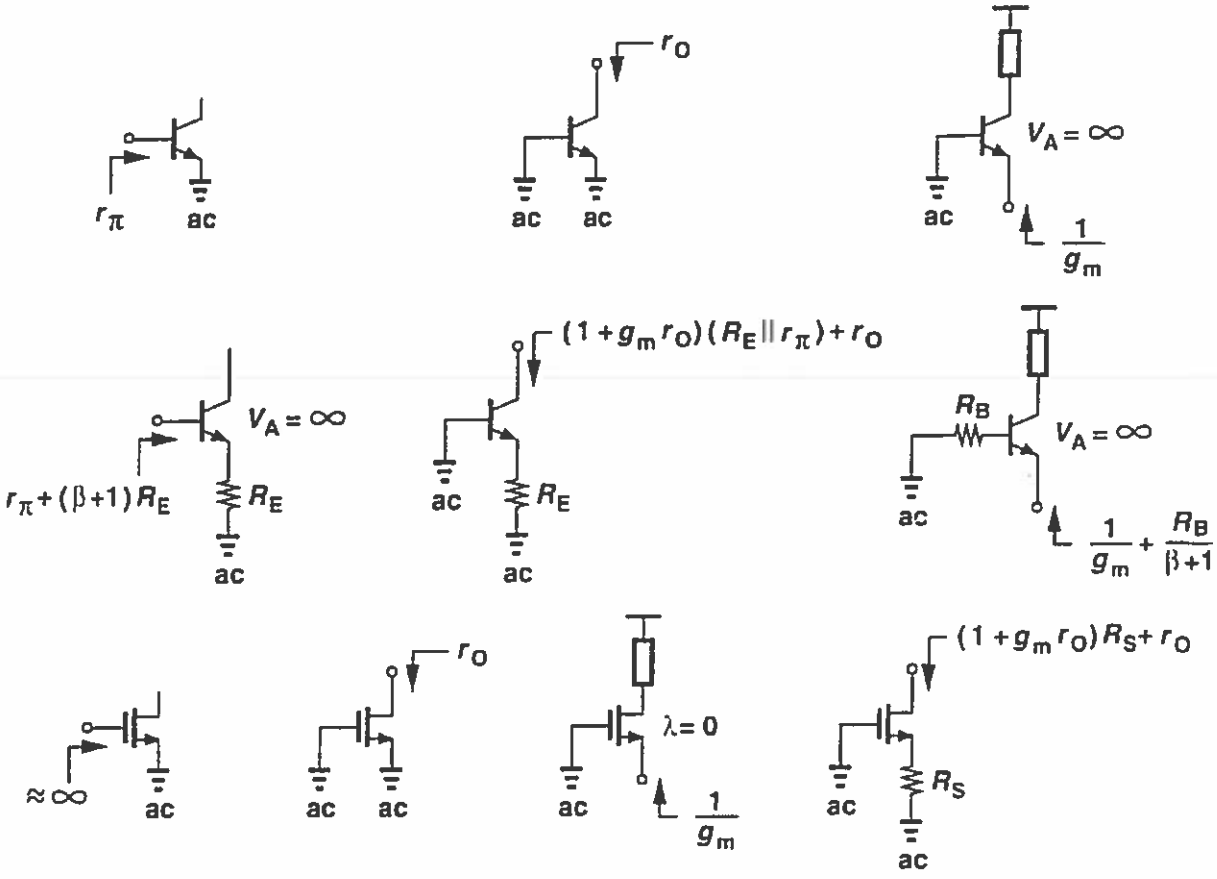
b) Using $A_v$=-$G_m R_{out}$, determine an expression for the *differential voltage gain*. [5]

c) Using your answer to part (b), calculate the differential voltage gain (in dB). [5]

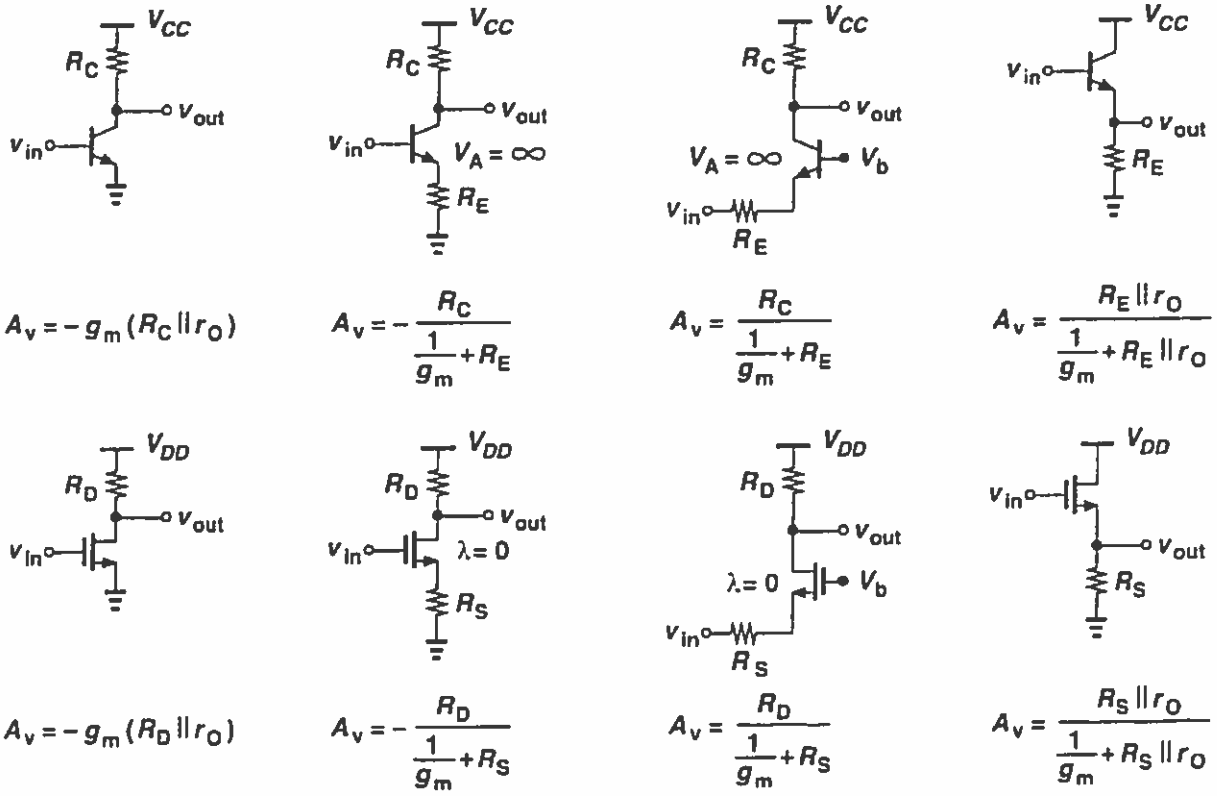
d) Determine expressions for the pole frequencies associated with nodes $X$ and $V_{out}$ (or equivalent in your half circuit). [5]

e) Evaluate your answer to part (d). [5]

## Input and Output Impedances




## Voltage Gain Equations





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